# Actual mode support and a fixed-family width test

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**Local result.** Accepted003/build003 elaborates12 statements, exit0/no warnings; four central printed dependencies contain only standard foundations. This draft is OUTSIDE Degree29's shared audit/replay. All records unsigned [pcc:unverified].

**Actual support.** Labels are $(r, b) \in \mathrm{Fin}(2^u) \times \mathrm{Fin}(k)$, $i < u$, with weight $w + b$. Nonzero polynomial left entries force $x \bmod 2^u = r$, hence row bit $i$ equals bit $i$ of $r$. Nonzero actual trimmed-right entries force the column bit to equal that same bit. With the original opposite column orientation, each label therefore has one fixed target side and one fixed source side. No arbitrary good-locality premise is supplied [pcc:unverified].

**Declared capacity.** At a fixed integer weight, projection to $r$ is injective, so at most $2^u$ labels have that weight. Across $P$ packets the bound is $P2^u$, even with different packet shifts $w$. For $i < u$,

$$P2^u = (P2^{i+1})2^{u-i-1}.$$

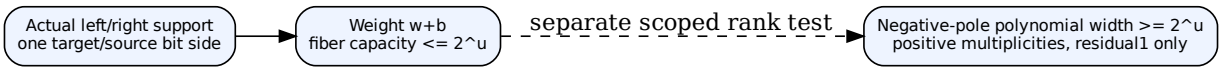
The comparison retains the weight shift; it is a declared-capacity identity, not equality with actual nonzero support or a runtime gain. The primary construction uses column tiles $2^{i+1}$ [pcc:unverified].

**Scoped polynomial width.** Take the actual normalized negative-pole chart matrix from the product family, $i < u \le a$, with all retained $m_j > 0$. For arbitrary rational-polynomial factors $A, B$ of inner width $R$, suppose

$$\mathrm{Content}_1(P^{\mathrm{neg}}_{xy} - (AB)_{xy}) \quad \text{for every } x, y.$$

Then $R \ge 2^u$. Evaluate at $Z = 1$: the actual chart is congruent to its normalized first jet; the prior exact class-minor theorem gives the width bound. No integrality premise on proposed $A, B$ is needed [pcc:unverified].

This concerns the fixed negative-pole product family at normalized residual1 under positive multiplicities. It is not an arbitrary-node, full-$W$, origin or general-algorithm lower bound, and does not prove that every $2^u k$ moment is necessary [pcc:unverified].




**Evidence and failures.** Exact source/log/exit pinned in local-validation-001.json. Three drafts/raw builds preserved. Failures included Boolean-not scope, Matroid operator tokenization, zero contradiction and a matrix-instance rewrite. The intended right-support equality was made explicit after the first failed parse; no accepted statement weakened. Accepted003 has no `sorryAx` or disabled linter [pcc:unverified].

Read primary D.12–16 and exact Rank40/FullChart/Degree/Compact APIs first; KB unavailable as journaled. Standard support/cardinality/rank algebra reused without novelty or speed claim. Child coefficient extraction, integral-ring transport, immutable creation precision, coefficient/bit/entry/recovery/ recursion/public expected cost remain separate. No attack, native solver, timing, paid or cloud campaign; all parent artifacts preserved.